PROJECT REPORT

MULTIMODAL AI APPLICATION

Audio Scene-to-Image Generation and Image Description-to-Speech Generation

A two-workflow application combining audio-text understanding, text-to-image synthesis, image captioning and text-to-speech

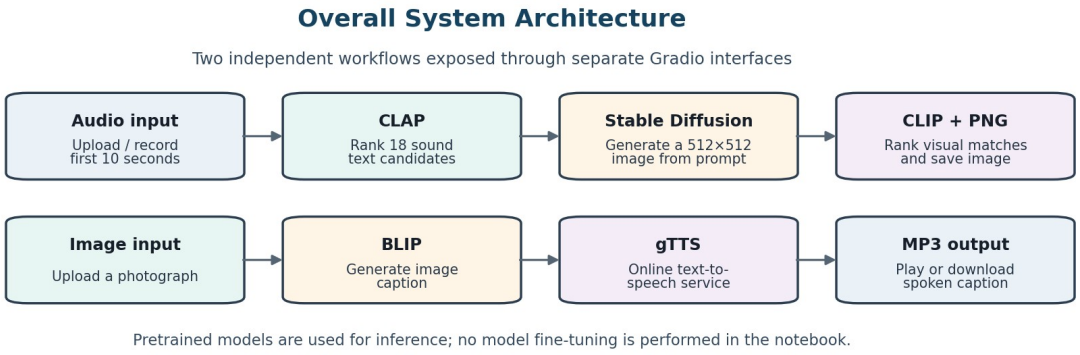


Figure 1. High-level architecture of the two application workflows.

Student Name: _____

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Department / Institution: _____

Course: Generative AI / Multimodal AI

Academic Year: 2026–2027

Prepared as an implementation and explanatory report

Report Scope and Reading Guide

This report documents the implementation in the supplied three-cell Google Colab notebook. It explains the two distinct workflows, the architecture and role of every model or service used, how data flows through the system, how to run and test the application, what can and cannot be concluded from the current demonstration, and answers to common project-viva questions.

> **Important implementation distinction**
>
> The project uses pretrained models and API services. The notebook does not train or fine-tune CLAP, Stable Diffusion, CLIP, or BLIP. gTTS is a client for an online text-to-speech service; the server-side speech model architecture is not exposed by this code.

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Terminology used in this report

Term	Meaning in this project
Embedding	A vector representation of audio, image, or text produced by a neural encoder.
Cosine similarity	A normalized measure of how aligned two embedding vectors are.
Zero-shot matching	Ranking candidate descriptions without retraining the model for each candidate label.
Diffusion	A generative process that learns to remove noise step by step.
Captioning	Generating a natural-language sentence that describes an image.
Text-to-speech (TTS)	Converting text into spoken audio.

1. Abstract and Keywords

This project implements a multimodal artificial-intelligence application with two complementary directions of conversion. In the first workflow, the user uploads or records environmental audio. Contrastive Language–Audio Pretraining (CLAP) compares the recording with a fixed set of 18 natural-language sound descriptions and selects the highest-ranked scene. That scene is mapped to a visual prompt, and Stable Diffusion v1.5 generates a 512 × 512 image. CLIP ViT-B/32 then compares the generated image with candidate visual descriptions and returns a ranking that serves as a semantic-alignment check.

In the second workflow, the user uploads a still image. BLIP image captioning generates a natural-language description, and gTTS converts that description to an English MP3 that can be played or downloaded from a Gradio interface. The implementation uses pretrained checkpoints and hosted text-to-speech functionality; it does not perform model training. The purpose is to demonstrate how specialised models can be integrated into end-to-end multimodal pipelines.

The application is designed for educational demonstration rather than high-stakes inference. The audio classifier is restricted to 18 candidate sound classes and analyses only the first 10 seconds of audio. The displayed similarity scores are relative to the candidate list and must not be interpreted as calibrated probabilities. Image captioning may omit objects or relationships, and generated images are plausible visual interpretations rather than reconstructions of the actual scene.

Keywords: Multimodal AI, CLAP, HTSAT, Stable Diffusion, latent diffusion, CLIP, BLIP, image captioning, text-to-speech, gTTS, Gradio, Hugging Face.

2. Introduction and Problem Definition

2.1 Background

Human perception combines multiple modalities. Sound conveys events and environmental context, while images convey objects, spatial arrangement, and appearance. Converting information between these modalities can improve accessibility, support creative applications, and make AI systems easier to demonstrate. Modern pretrained models provide specialised capabilities for matching sound to text, generating images from prompts, aligning images with captions, describing images, and synthesising speech.

2.2 Problem statement

A user needs a simple interface that demonstrates two useful multimodal tasks: (A) convert an environmental sound into a generated visual representation and (B) convert an uploaded image into a spoken description. Implementing each capability from scratch would require large datasets, training infrastructure, and substantial model-development effort. The project therefore integrates pretrained models into a single notebook while keeping the two processing pipelines understandable and testable.

2.3 Motivation

- **Accessibility:** spoken captions can provide an audio alternative to visual descriptions for users who prefer listening or have difficulty reading a caption.

- **Creative exploration:** environmental audio can inspire a visual interpretation even when no image of the original scene is available.
- **Modular AI design:** specialised pretrained models can be connected through explicit intermediate representations such as a scene label or caption.
- **Interactive learning:** the Gradio interface makes model inputs, outputs, prompts, and candidate rankings visible to the user.

2.4 Intended use

The project is intended for coursework, demonstrations, and experimentation. It is not a validated accessibility product, a surveillance system, a sound-event benchmark, or a guaranteed-faithful audio-to-image reconstruction tool.

3. Objectives and Scope

3.1 Objectives

- Build an audio-to-image interface that accepts uploaded or recorded sound.
- Rank 18 candidate scene descriptions against an audio embedding using CLAP.
- Map the selected scene to a text prompt and generate an image using Stable Diffusion v1.5.
- Use CLIP ViT-B/32 to rank the generated image against candidate visual scene descriptions.
- Build an image-to-speech interface that captions an uploaded image using BLIP.
- Convert the caption to an English MP3 using gTTS and expose playback/download controls.
- Document model architecture, parameters, expected behaviour, limitations, and viva questions.

3.2 Scope boundaries

Included	Not included in the current notebook
Pretrained inference for both workflows	Training or fine-tuning the pretrained models
18 predefined environmental-sound categories	Open-ended sound event discovery across any sound class
512 × 512 prompt-based image generation	Reconstruction of the real scene from sound; frame-accurate or physically verified imagery
CLIP candidate-scene ranking	Calibrated probability estimation or objective image-quality scoring
BLIP image captioning and gTTS speech output	Offline neural speech synthesis or multi-language UI controls
Gradio-based prototype interface	Production authentication, monitoring, deployment hardening, or formal accessibility certification

4. System Requirements and Technology Stack

4.1 Software and runtime

Component	Role in the project
Google Colab / Python	Notebook execution and dependency environment.
PyTorch	Tensor operations and neural-network inference.
Transformers	Loads CLAP, CLIP and BLIP processors and pretrained checkpoints.
Diffusers	Loads and runs the Stable Diffusion text-to-image pipeline.
librosa	Loads, resamples and converts audio to mono for CLAP inference.
Pillow (PIL)	Image decoding, RGB conversion and saving generated PNG output.
NumPy / pandas	Embedding-score processing and readable top-ranked tables.
Gradio	Browser UI for upload, microphone input, generated output, audio playback and download.
gTTS	Sends caption text to an online text-to-speech service and saves MP3 audio.

4.2 Hardware

A CUDA-enabled Colab GPU, such as a T4, is recommended because Stable Diffusion performs iterative denoising and is substantially slower on CPU. In the supplied notebook, CLAP and the separate CLIP evaluator are placed on the CPU in FP32 to avoid the earlier CUDA FloatTensor/HalfTensor batch-normalisation error and to conserve GPU memory. Stable Diffusion uses FP16 on CUDA when available, and FP32 if it falls back to CPU. BLIP is loaded in FP32 onto the available compute device. Since these model weights can consume substantial memory, a high-memory GPU runtime is preferred.

4.3 Network and model downloads

The first run downloads model weights and tokenizer/processor files from Hugging Face. The gTTS step requires a working internet connection when speech is generated because the notebook uses an online client rather than a locally loaded speech checkpoint. If Colab requests a runtime restart after dependency installation, restart once and then execute the cells in order.

5. Overall System Architecture

The notebook contains two separate Gradio applications. They share a runtime but do not form a single serial pipeline: audio-to-image does not call BLIP or gTTS, and image-to-speech does not call CLAP or Stable Diffusion. The workflows have independent inputs, intermediate representations, model calls, and outputs.

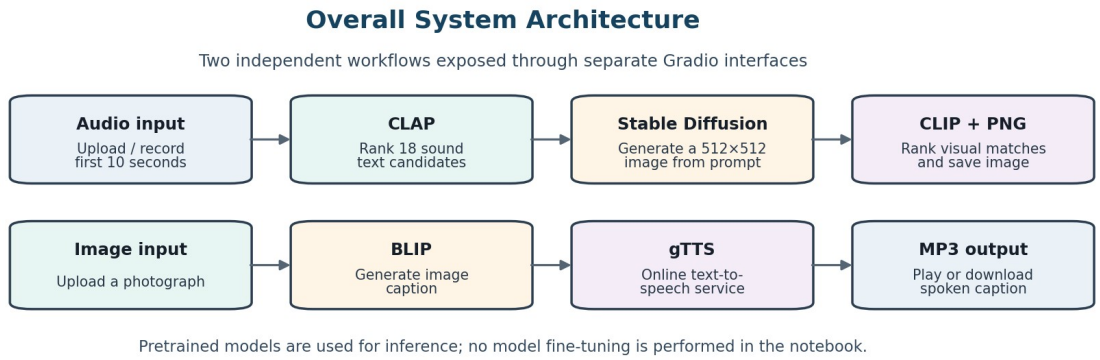


Figure 2. The two independent multimodal workflows and their major transformations.

5.1 Workflow A: audio to image

1. Audio input: the user uploads a recording or records through the browser microphone.
2. Preprocessing: librosa reads the first 10 seconds, converts to mono, and resamples to 48 kHz.
3. Scene matching: CLAP creates an audio embedding and compares it with cached embeddings of 18 candidate sound descriptions.
4. Prompt construction: the top-ranked scene is mapped to a predefined visual description; the user may override it with a custom prompt.
5. Image synthesis: Stable Diffusion denoises a randomly initialised latent representation and decodes the result to a 512 × 512 image.
6. Semantic check: CLIP scores the generated image against 18 candidate visual descriptions; the app presents the top five matches.
7. Output: the generated image and PNG download are displayed with the predicted scene and ranking tables.

5.2 Workflow B: image to speech

1. Image input: the user uploads a still image.
2. Image preprocessing: PIL converts the image to RGB, and the BLIP processor prepares the model inputs.
3. Caption generation: BLIP produces a sentence conditioned on visual features.
4. Speech generation: gTTS sends the caption to an online text-to-speech service and writes an MP3 file.
5. Output: Gradio displays an editable caption, an audio player and a download link. The user may edit the caption and regenerate speech.

6. Workflow A: Audio Scene-to-Image Generation

6.1 Input handling and audio preprocessing

The app accepts an uploaded or recorded audio file through Gradio. librosa loads the first 10.0 seconds, converts it to mono, and resamples it to 48,000 samples per second. The code rejects missing, undecodable, empty, or nearly silent recordings. The first-10-second limit bounds inference time, but it also means sounds occurring later in a longer recording do not influence the predicted scene.

6.2 CLAP-based scene ranking

CLAP means Contrastive Language–Audio Pretraining. It is a dual-encoder model trained to bring the representations of corresponding audio clips and text descriptions close together in a shared embedding space [1, 2]. The selected checkpoint, `laion/clap-htsat-unfused`, uses an HTSAT-based audio encoder. The text branch is a Transformer text encoder. At inference, the project encodes 18 prewritten sound descriptions once, caches their text embeddings, and computes the audio embedding for each input recording.

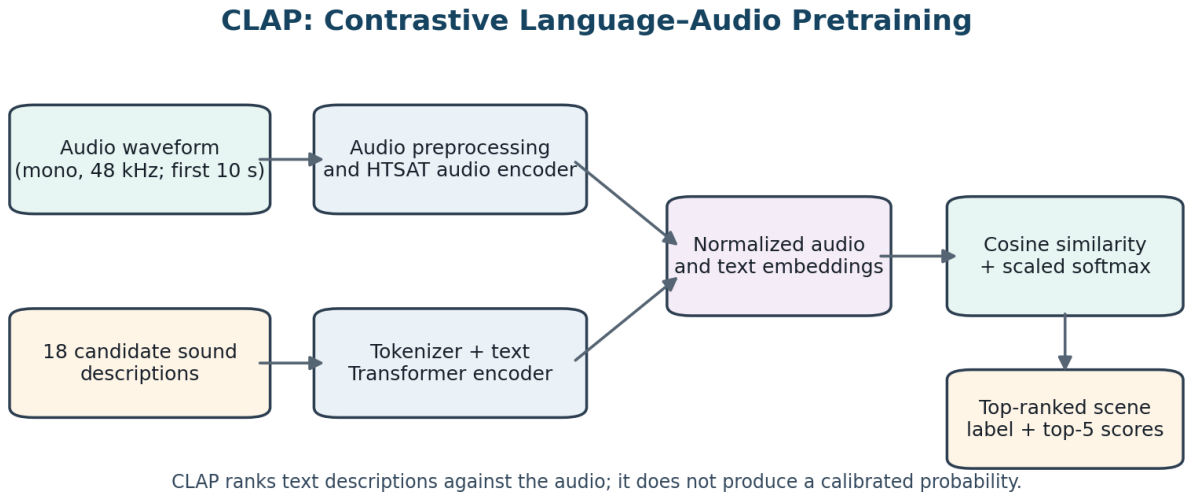


Figure 3. Project-level view of the CLAP inference path. The candidate text embeddings are cached to reduce repeat work.

6.3 Similarity scores and predicted category

Both the audio vector and each text vector are L2-normalised. Their matrix product is therefore a cosine-similarity-like score. The implementation applies a scale factor of 10 and a softmax to obtain a ranking distribution over the current 18 candidates. The candidate with the largest value becomes the predicted scene; the top five values are shown in a table.

> **How to explain the displayed score**
>
> A value of 62% in the table would mean that this candidate received 62% of the softmax-normalised score mass for this particular candidate set and scaling. It does not mean that the scene is correct with 62% real-world probability. Adding or changing candidate descriptions may change the percentages.

6.4 Scene-to-prompt mapping

The project has a dictionary of 18 classes, each paired with an audio description and a visual description. Examples include Rain, Thunderstorm, Ocean Waves, Birds, Forest, Traffic, Train, Aircraft, Dog, Cat, Fire, Wind, People Talking, Applause, Footsteps, Cooking, Machinery, and Music. For example, the Thunderstorm label is mapped to a prompt describing a dark stormy sky, lightning and heavy rain. A custom prompt can replace the predefined visual prompt, although the predicted label is still reported.

6.5 Stable Diffusion image synthesis

Stable Diffusion v1.5 is a latent text-to-image diffusion model [3, 4]. The app uses the Hugging Face Diffusers `StableDiffusionPipeline`, generates a 512 × 512 image, allows 10–30 inference steps (default 20), uses classifier-free guidance scale 7.0, and accepts a random seed (default 42). The negative prompt discourages blur, low quality, distortion, artifacts, watermarks, and unreadable text. An explicit seed makes runs more repeatable under the same software, hardware and settings, but identical output is not guaranteed across all runtimes or library versions.

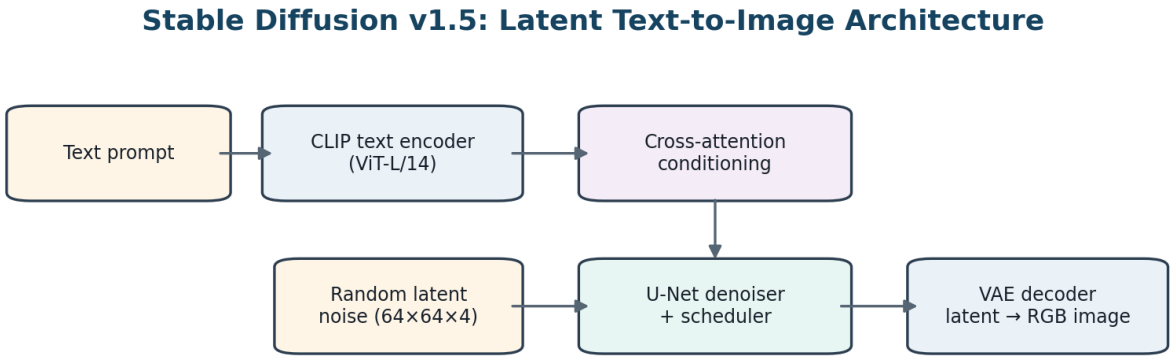


Figure 4. Main Stable Diffusion v1.5 inference components. The text encoder conditions the U-Net through cross-attention.

6.6 CLIP-based generated-image evaluation

CLIP stands for Contrastive Language–Image Pre-training. The checkpoint `openai/clip-vit-base-patch32` has a ViT-B/32 image encoder and a Transformer text encoder. The encoders map images and text into aligned embedding spaces trained with a contrastive objective [5, 6]. After image generation, the project compares the generated image embedding with embeddings of the 18 candidate visual descriptions and displays the five highest-scoring matches.

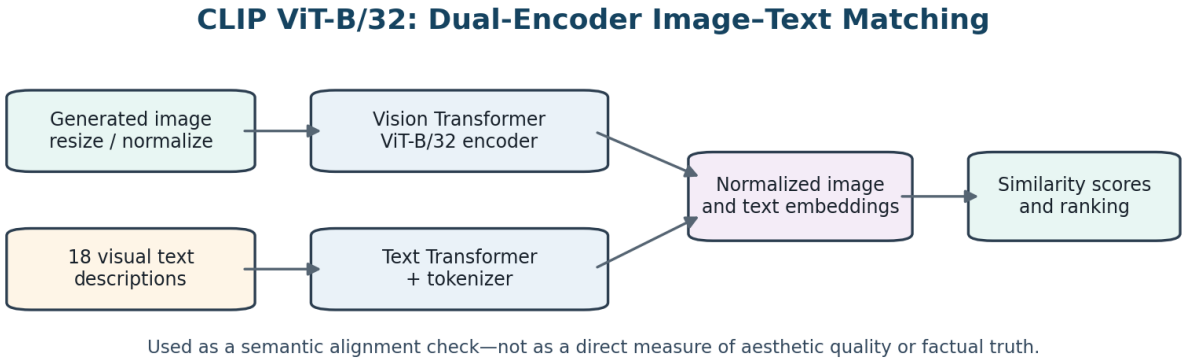


Figure 5. CLIP is used as a semantic-matching check between the generated image and candidate text descriptions.

CLIP does not determine whether an image is aesthetically pleasing, factually true, or a faithful reconstruction of the original scene. Its ranking tells the user which descriptions align most closely with the generated image according to the model. It is not a substitute for a human review or a calibrated quality metric.

7. Workflow B: Image Description-to-Speech Generation

7.1 Image input and preprocessing

The second Gradio app accepts a PIL image. The image is converted to RGB before passing it through the BLIP processor, which performs the model-specific resizing, normalisation, and tensor formatting. If no image is provided, the app returns a clear interface error instead of attempting inference.

7.2 BLIP image captioning

BLIP stands for Bootstrapping Language–Image Pre-training. The selected checkpoint is `Salesforce/blip-image-captioning-base`, described on its model card as a base architecture with a Vision Transformer backbone, pretrained for image captioning [7, 8]. In this application, the image encoder extracts visual features and the image-conditioned text decoder generates a caption token by token. The code uses beam search with five beams, limits output to 60 new tokens, and sets a repetition penalty of 1.15. These are decoding settings, not training parameters.

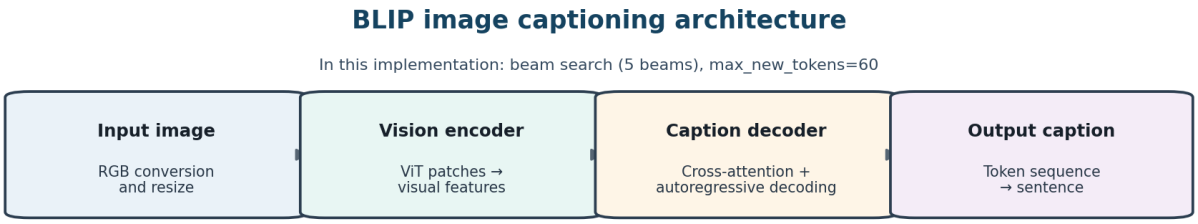


Figure 6. BLIP transforms image pixels into visual features and generates a caption conditioned on those features.

7.3 gTTS text-to-speech

The generated caption is sent to `gTTS`, a Python client for an online text-to-speech service. The code requests English speech (`lang="en"`, normal speed) and saves it as an MP3 file [9]. The project does not download or host the internal neural TTS model, so the report should not attribute an undocumented neural-network architecture to gTTS. The online request also means speech generation depends on internet connectivity and on service availability.

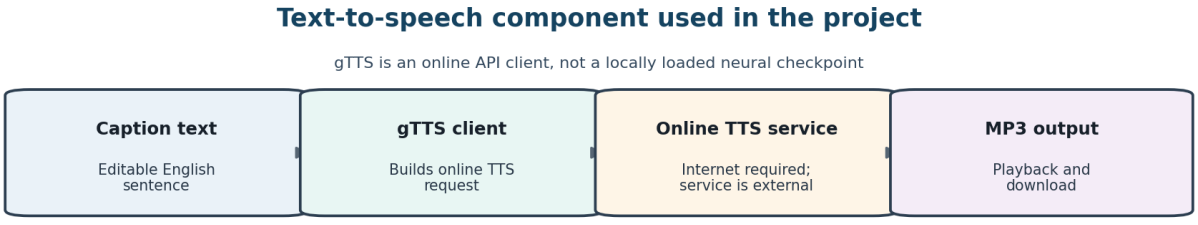


Figure 7. The gTTS component sends text to an online service; the response is stored as MP3 audio.

7.4 Editing and regenerating speech

The caption appears in an editable textbox. This is useful when the model produces an incomplete or awkward description: the user can correct the sentence and click the separate speech-generation button to create a new MP3 without rerunning BLIP. The interface provides audio playback and a download link.

8. Model Architecture Details: At a Glance

Model / component	Architecture or mechanism	Input → output	Role here
CLAP — laion/clap-htsat-unfused	Dual encoder: HTSAT audio encoder + Transformer text encoder; contrastive audio/text embeddings.	Audio waveform + text candidates → aligned vectors / ranking	Selects one of 18 sound-scene candidates.
Stable Diffusion v1.5	Latent diffusion: CLIP ViT-L/14 text encoder, conditional U-Net, scheduler, VAE decoder.	Text prompt + random latent noise → RGB image	Generates the scene illustration.
CLIP ViT-B/32	Dual encoder: ViT-B/32 visual Transformer + text Transformer; contrastive alignment.	Generated image + candidate descriptions → ranking	Checks semantic alignment of generated output.
BLIP image-captioning base	Vision Transformer encoder + image-conditioned autoregressive language decoder.	RGB image → caption tokens / sentence	Describes uploaded images.
gTTS	Online TTS client; neural service internals are not exposed in the project code.	Caption text → MP3 audio	Speaks and exports the image caption.
Gradio	Python UI/event framework; not a machine-learning model.	User events → model functions → rendered outputs	Connects model functions to browser controls.

8.1 Why there are multiple vision-language models

The system uses CLIP and BLIP for different purposes. CLIP is a dual encoder: it creates embeddings and scores alignment between an image and candidate text descriptions. BLIP is used for generation: it conditions a text decoder on image features to produce a fluent caption. Stable Diffusion also contains its own CLIP text encoder (ViT-L/14) for prompt conditioning. That internal text encoder is not the separately loaded `openai/clip-vit-base-patch32` model used to rank the generated image.

8.2 Model usage and licensing checks

Model cards and licences can change or include important usage restrictions. The notebook lists the BLIP model card as BSD-3-Clause and the Stable Diffusion v1.5 card as CreativeML OpenRAIL-M; users should read the corresponding terms before redistribution or commercial use [4, 7]. This report does not provide legal advice. The license of a model checkpoint, the license of its source code, and the terms of an online service are not necessarily the same.

9. Implementation Details and Important Parameters

Parameter	Value in supplied notebook	Explanation
Audio duration analysed	First 10 seconds	Bounds runtime; later events are ignored.
Audio sample rate	48,000 Hz; mono	Matches the preprocessing expectation for the selected CLAP pipeline.
Number of audio candidates	18	Predictions are limited to the predefined scene dictionary.
CLAP runtime dtype/device	FP32 on CPU	Avoids the previous CUDA FP32/FP16 batch-normalisation error.
CLIP runtime dtype/device	FP32 on CPU	Reduces GPU memory competition and uses matching precision.
Stable Diffusion resolution	512 × 512	Output image size used in the notebook.
Stable Diffusion steps	10–30; default 20	More steps can improve refinement but increase latency; not a guarantee of quality.
Guidance scale	7.0	Controls influence of prompt conditioning.
Seed	Default 42; user adjustable	Controls initial random latent noise for repeatability.
BLIP decoding	5 beams; max 60 new tokens; repetition penalty 1.15	Controls caption search, length and repeated text.
TTS output	English MP3; normal speed	Uses gTTS with an internet connection.
User interface	Two Gradio apps	One interface for audio-to-image and one for image-to-speech.

9.1 Dependency installation and startup

1. Run notebook Cell 1 to uninstall optional `torchao` and install the listed dependencies while preserving the Colab PyTorch build.
2. If Colab prompts for a runtime restart, restart once after installation.
3. Run Cell 2 to load CLAP, cache candidate text embeddings, load the CLIP evaluator and Stable Diffusion, then launch the audio-to-image Gradio interface.
4. Run Cell 3 to load BLIP and launch the image-to-speech Gradio interface.
5. If the runtime runs out of memory, restart and test one app at a time or use a higher-memory GPU. Do not assume two model-heavy applications will fit on every runtime.

9.2 Error handling

The code checks for missing audio/images, empty waveforms, and near-silent sound. Exceptions are wrapped in Gradio errors so they appear in the interface. If gTTS fails, the first checks should be internet connectivity and service availability. If a tensor dtype error reappears, verify that CLAP and

CLIP are actually loaded in FP32 and the audio feature tensors are passed to the CPU model without unintended device/precision conversion.

10. Testing, Demonstration and Evaluation

10.1 Functional test plan

Test ID	Input / condition	Expected behaviour	Evidence / status
T1	Clear environmental audio (e.g., thunderstorm)	CLAP ranks candidate descriptions; top label is displayed; an image is generated.	A supplied screenshot shows “Thunderstorm” and a generated storm image; qualitative demonstration only.
T2	Rain / traffic / birds audio	Model produces a ranking among the 18 classes.	To be checked with representative recordings.
T3	No audio selected	App shows a validation error.	Behaviour specified in code; verify in runtime.
T4	Silent or undecodable audio	App rejects empty or near-silent waveform.	Behaviour specified in code; verify in runtime.
T5	Custom prompt supplied	Custom prompt replaces default visual prompt for image generation.	Behaviour specified in code; verify in runtime.
T6	Generated image evaluation	CLIP top-five candidate visual matches are returned.	Ranking is semantic alignment, not calibrated probability.
T7	Upload a clear photo	BLIP returns an editable caption.	To be checked in the live app.
T8	Generate speech from caption	gTTS creates an MP3 for playback/download.	Requires internet; to be checked in the live app.
T9	Blank caption / no image	UI raises validation error.	Behaviour specified in code; verify in runtime.
T10	No internet during speech generation	gTTS reports a useful error and output cannot be created.	Expected dependency behaviour.

10.2 Demonstration screenshot

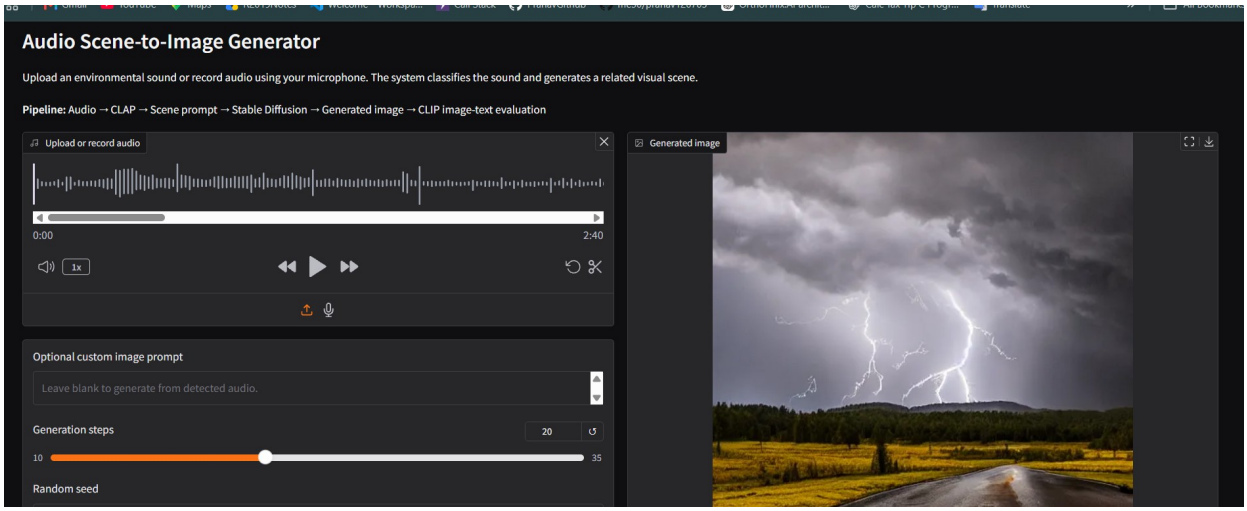


Figure 8. Supplied Gradio demonstration: an audio clip was assigned the scene "Thunderstorm" and a storm-related image was generated.

The screenshot is evidence that the audio-to-image interface was demonstrated for one recording. It is not a quantitative evaluation and it does not establish accuracy across all 18 sound categories. The 2-minute-40-second player duration visible in the screenshot also does not change the code-level fact that classification analyses only the first 10 seconds of an input recording.

10.3 Evaluation methodology recommended for a formal assessment

To report measured performance, assemble a labelled test set with multiple recordings per sound category, including difficult cases and background noise. Run each sample through the unchanged pipeline and compare the top-ranked CLAP label with its ground-truth class. Report accuracy, per-class precision/recall/F1, and a confusion matrix. For image outputs, conduct human evaluation or use task-specific image-text alignment and visual quality metrics, clearly stating their limitations. For captioning, compare descriptions with human-written references using metrics such as CIDEr or SPICE only if the evaluation setup and reference captions are properly defined. For TTS, confirm intelligibility, language, and duration through listening tests. No such measured benchmark is included in the supplied notebook; therefore, this report does not invent accuracy or quality percentages.

11. Limitations, Responsible Use and Future Work

11.1 Current limitations

- **Closed candidate set:** CLAP can only choose among the 18 provided descriptions for this application. A sound not represented by these descriptions will still be assigned a best match.
- **Short audio window:** only the first 10 seconds are analysed, so important events later in the clip are ignored.
- **Ambiguous and mixed audio:** background music, speech, traffic and weather can occur together; the single-label output may oversimplify a scene.
- **Prompt mapping:** the scene-to-visual dictionary is manually written. The mapping expresses a plausible interpretation, not a verified image of the original event.
- **Generation variability:** Stable Diffusion may add, omit or distort details. A fixed seed does not guarantee identical outputs across all environments.
- **Relative scores:** CLAP and CLIP softmax values depend on the candidate list and temperature/scale; they are not calibrated confidence probabilities.
- **Captioning errors:** BLIP can miss small objects, count incorrectly, or describe an action/relationship not clearly supported by the pixels.
- **Online TTS dependency:** gTTS requires internet access and reliance on an external service; it is not fully offline.
- **Memory / latency:** model downloads, inference latency and GPU memory requirements can make the application slow or cause out-of-memory errors.
- **Model and data bias:** the pretrained models may reflect biases and limitations in their training data; outputs need human review.

11.2 Responsible use

Do not treat generated images as documentary evidence of what happened in an audio recording. Do not rely on automatic captions for safety-critical decisions without human verification. Avoid uploading sensitive personal images or private recordings to a hosted notebook or third-party service without considering privacy and service terms. Review model-specific licences and online-service terms before public or commercial deployment.

11.3 Future improvements

- Analyse long audio using overlapping chunks and aggregate candidate scores over time.
- Allow user-defined sound descriptions and show the margin between the top candidate and alternatives.
- Add a learned or LLM-based prompt-expansion step, while making clear that expanded details may be invented.
- Add a confidence calibration/abstention policy using a labelled validation dataset.
- Include a feedback button to record incorrect scene labels and user ratings for generated images and captions.
- Add a local/offline TTS engine and selectable languages/voices.
- Add evaluation datasets, repeatable test scripts, latency measurements, resource profiling and version logging.
- Improve deployment safety: remove public-sharing links where not needed, add access controls for real deployments, and make temporary-file cleanup explicit.

12. Conclusion

The project demonstrates two practical multimodal AI pipelines using pretrained models: environmental audio is mapped to a candidate scene through CLAP, turned into an image through Stable Diffusion, and checked against candidate descriptions through CLIP; an uploaded image is converted to a natural-language caption through BLIP and read aloud through gTTS. Gradio connects these stages to accessible browser interfaces.

The main engineering contribution is the integration of distinct model families with explicit intermediate representations, runtime configuration, and error handling rather than training a new foundation model. The model architecture and limitation distinctions are important when explaining the project: CLAP ranks a closed set of sound descriptions, Stable Diffusion creates a plausible visual output, CLIP estimates semantic match, BLIP generates a caption, and gTTS depends on an online speech service. A formal quality claim requires a defined test set and actual measurements; the current demonstration should be presented as a working prototype, not as a quantitatively validated system.

13. Viva / Review Questions and Answers

1. What is the project about?

It is a multimodal AI prototype with two workflows: environmental audio to a generated image, and an uploaded image to a spoken caption.

2. What does multimodal mean?

It means processing or connecting different data modalities. This project uses audio, images and text, and produces either images or speech.

3. Why are there two separate applications?

Their inputs and pipelines differ. The audio-to-image app needs CLAP, Stable Diffusion and CLIP; the image-to-speech app needs BLIP and gTTS.

4. What is CLAP?

Contrastive Language–Audio Pretraining. It learns aligned audio and text embeddings so an audio clip can be matched to natural-language descriptions without training a classifier for every new label.

5. What is HTSAT?

HTSAT is an audio encoder architecture based on a hierarchical Swin Transformer over audio spectrogram representations. In this system it produces an embedding for audio-text matching.

6. Does CLAP directly generate a sentence describing the audio?

Not in this notebook. It ranks 18 supplied descriptions and returns the top candidate scene label. It is not configured here as a free-form audio caption generator.

7. Why does the code use 18 sound classes?

The fixed dictionary makes the demonstration manageable and lets the model compare each recording with a known set of candidate descriptions. It limits coverage of unseen sounds.

8. How is the audio preprocessed?

librosa loads the first 10 seconds, converts the signal to mono, and resamples to 48 kHz before passing it to the CLAP processor.

9. How is the predicted category selected?

The audio embedding is compared with cached text embeddings. The code computes normalised similarities, scales them by 10, applies softmax and chooses the highest-ranked candidate.

10. Are the CLAP scores real-world probabilities?

No. They are softmax-normalised relative scores over the chosen candidate set. They are not calibrated against labelled validation data.

11. What is Stable Diffusion?

A latent diffusion text-to-image model. It generates an image by iteratively denoising a compressed latent representation conditioned on text embeddings.

12. What are the three core Stable Diffusion components?

A text encoder, a conditional U-Net denoiser and a variational autoencoder (VAE), together with a scheduler that updates the latent during denoising.

13. Why use latent diffusion instead of diffusion directly on pixels?

The denoising process occurs in a smaller latent space, reducing compute compared with running the same process directly over full-resolution pixels.

14. What is classifier-free guidance?

A sampling technique that combines conditional and unconditional noise predictions to adjust how strongly the generated image follows the prompt. The project uses guidance scale 7.0.

15. What does the random seed do?

It controls the starting random noise for generation. With the same seed and environment, it improves repeatability; it does not guarantee identical output across versions or hardware.

16. Why use CLIP after generating an image?

CLIP embeds the generated image and candidate scene texts in a joint space, then ranks the descriptions by similarity. It is a semantic alignment check, not a direct measure of image quality.

17. Is the CLIP model used for evaluation the same CLIP inside Stable Diffusion?

No. Stable Diffusion uses an internal CLIP ViT-L/14 text encoder for prompt conditioning. The notebook separately loads OpenAI CLIP ViT-B/32 to compare completed images against candidate descriptions.

18. What is BLIP?

Bootstrapping Language–Image Pre-training. The project uses a pretrained image-captioning version with a vision encoder and an image-conditioned text decoder to generate a sentence.

19. How does BLIP generate the caption?

The image is processed into visual features. A language decoder generates caption tokens conditioned on those features; beam search with five beams is used in the supplied implementation.

20. What is gTTS and why is it used?

gTTS is a Python client that calls an online text-to-speech service and writes MP3 speech. It is lightweight to integrate, but requires internet and does not expose the service's internal model architecture.

21. Why is the caption editable?

Captioning models may omit details or use awkward wording. Editing lets the user correct the sentence and regenerate speech without rerunning the image model.

22. What is Gradio?

A Python library for building a web UI around model functions. It provides the file inputs, buttons, generated outputs, audio player and download controls.

23. Did the project train any models?

No. It integrates pretrained checkpoints and an online TTS service. The notebook performs inference only.

24. What happens if the sound is not one of the 18 categories?

The classifier still returns the highest-scoring candidate. That result can be wrong because the candidate set does not include every possible sound.

25. What are the major limitations?

Closed-set sound ranking, a 10-second audio window, ambiguous mixed sound, prompt-based image invention, captioning errors, uncalibrated similarity scores, internet dependence for gTTS and runtime/memory constraints.

26. How would you evaluate the audio classifier properly?

Use a labelled test set, compare the top predicted class with ground truth, and report accuracy, per-class precision/recall/F1 and a confusion matrix. None of these results should be invented without running the evaluation.

27. How would you evaluate image-caption quality?

Collect reference captions and use suitable caption metrics such as CIDEr or SPICE alongside human evaluation, with the dataset and protocol clearly specified.

28. Why did an FP32/FP16 CUDA error occur earlier?

A model input tensor and its layer weights had different dtypes during batch normalisation. The notebook avoids that specific mismatch by running CLAP and the separate CLIP evaluator in FP32 on CPU.

29. What are the hardware constraints?

Stable Diffusion needs the most compute and GPU memory because it runs a sequence of denoising steps. Running BLIP in the same session can add memory pressure.

30. What future improvements would you propose?

Add a broader sound-label set, long-audio aggregation, confidence calibration, robust benchmarks, user feedback, local TTS, multilingual support and better resource/deployment management.

31. Can all models be implemented and trained from scratch for this project?

Writing the architectures from scratch is possible, but training CLAP, Stable Diffusion and BLIP to useful quality by tomorrow is not realistic because they rely on large-scale training data. gTTS is an online service, not a model trained in this notebook. This project uses pretrained inference and does not claim model training. A reduced-scope CNN trained on a small sound dataset is possible, but its coverage and output quality would be limited.

14. References and Model Resources

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Appendix A. One-minute verbal explanation

“My project is a multimodal AI application with two independent workflows. First, it takes environmental audio, resamples the first ten seconds to mono 48 kHz, and uses CLAP to rank eighteen natural-language sound descriptions. The top label selects a visual prompt. Stable Diffusion v1.5 generates a 512-by-512 image from that prompt, and CLIP ViT-B/32 compares the generated image with candidate scene descriptions as a semantic check. Second, the user uploads a photograph; BLIP generates a caption from its visual features, and gTTS converts the caption into a playable MP3. Gradio provides the interfaces. All neural checkpoints are pretrained; no fine-tuning was performed. The main limitations are the fixed sound categories, possible caption/generation errors, relative rather than calibrated scores, and the internet requirement for gTTS.”

Appendix B. Suggested demonstration sequence

1. Demonstrate audio-to-image first with a clear 5–10 second sound sample and explain that only the first 10 seconds are analysed.
2. Read the top five CLAP rows, identify the predicted scene and explain why scores are relative candidate rankings.
3. Show how changing the custom prompt changes the image while leaving the sound-classification step unchanged.
4. Point out that CLIP is a semantic consistency check and not an image-quality score.
5. Switch to the image-to-speech app, upload a clear photograph, read the BLIP caption, edit it if necessary, and generate/download the MP3.
6. Conclude with limitations and state that no formal accuracy benchmark was measured in the submitted notebook.

Appendix C. Model quick reference for the presentation

Component	Say this in one sentence	Avoid claiming
CLAP / HTSAT	Ranks the provided text descriptions against an audio embedding.	It does not freely describe all possible sounds or provide calibrated probabilities.
Stable Diffusion v1.5	Generates a new image by denoising latent noise under text conditioning.	The image is not evidence of the exact real-world scene.
CLIP ViT-B/32	Ranks how well a generated image aligns with candidate texts.	The score is not a direct image-quality metric.
BLIP captioning	Generates a sentence conditioned on the uploaded image.	A caption can omit details or be mistaken.
gTTS	Uses an online TTS service client to create an MP3 from the caption.	It is not an offline model packaged in the notebook.